The Value of Targeting Gritting with Phone Vibration Detections

The mathematical model of the slipperiness simulator, with proofs

Elias Wuolijoki

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Abstract

We study a simulator in which pedestrians' phones detect slips from vibration and gritting is directed by these detections. The walkable squares of central Turku have a hidden slipperiness that changes daily and that nobody observes directly. The maintenance crew can grit only a few blocks at a time, and a block may not be gritted again before the effect of its previous gritting has ended.

We prove that targeting based on the beta-binomial posterior computed from the detections yields, in expectation, at most as many falls as any policy that ignores the detections. Compared with the best such policy, the improvement is strictly positive exactly when slipperiness varies between squares and the detector distinguishes slips from false alarms. The results concern the simulated world, not real streets in Turku or If's claims.

Contents

1 Introduction **2**

2 The simulated world **3**

3 What the service knows: the beta-binomial posterior **8**

4 Decomposing the value of gritting **10**

5 Main results **11**

6 Correspondence between code and model **15**

7 Numerical results **16**

8 Limitations and further work **17**

1 Introduction

1.1 The question

The service idea is simple. When a pedestrian slips, the phone's accelerometer notices it. If such detections accumulate at one place, the place is probably slippery, and that is where gritting should go.

The idea sounds obvious, but it is not. Detections are rare and noisy, and some of them are false alarms. A busy street collects detections simply because many people walk there. Moreover, a crew could just as well rotate through the blocks in order or always grit the busiest streets. So the question is:

Are vibration detections genuinely useful compared with the best way to grit without them?

1.2 The answer in brief

Let $J(\pi)$ denote the expected number of falls in January 2026 under policy π. Under the assumptions of Section 5 we prove the chain

$$J(\texttt{oracle}) \;\le\; J(\texttt{pito}) \;<\; J(\texttt{busiest}) \;\le\; J(\texttt{rotation}) \;\le\; J(\texttt{no-grit}).$$

The policies in the chain are as follows. `pito` uses the vibration detections, and `busiest` grits the busiest blocks. `rotation` goes through the blocks in order, and `oracle` knows the hidden slipperiness. The most important part of the chain is the strict inequality in the middle: by Theorem 1, `busiest` is the best possible policy that does not use detections. The detections therefore beat *every* rule that ignores them, not merely an arbitrarily chosen benchmark.

By Theorem 5 the conditions are also necessary: if slipperiness does not vary between squares, or if the detector cannot tell slips from false alarms, the gain is exactly zero.

1.3 What this paper does not claim

The theorems concern a simulated world in which the service knows the structure and parameters of the model. They do not show that Turku's pavements follow this model, that gritting reduces slips by 80 per cent, or that a phone detects slips with the stated accuracy. The theorems give the ordering; the *size* of the differences depends on the assumptions and is estimated by simulation in Section 7.

1.4 How to read this paper

Section 2 describes the simulated world. Section 3 shows how the service infers slipperiness from detections. Section 4 turns the value of gritting into one selection problem per gritting round, and Section 5 proves the main results. Sections 6 and 7 connect the results to the code and to numbers.

A reader in a hurry can read the introduction, Figure 1, the theorems of Section 5 with their intuition boxes, and Table 4.

2 The simulated world

2.1 Overview

Figure 1 shows what happens in one square during one hour. Grey boxes are the truth that the simulator keeps hidden; blue boxes are what the service sees. The service never sees a square's slipperiness or the falls, only passages and vibration detections.

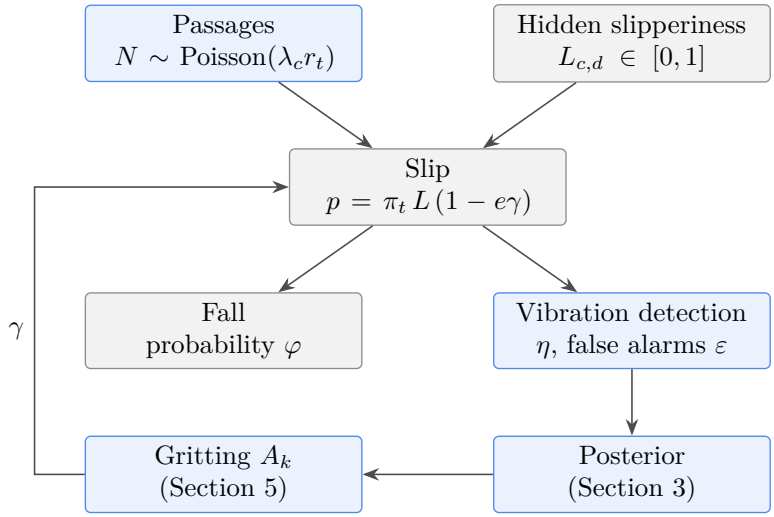


Figure 1: Structure of the simulator in one square and one hour. Grey: hidden truth. Blue: what the service sees and the decisions it makes. Gritting γ reduces the slip risk in the following hours.

2.2 Time and space

Time advances in hours, $t \in \mathcal{T} = \{1, \ldots, T\}$, where $T = 744$ is the number of hours in January 2026. Hour $t = 1$ is 00–01 on 1 January 2026, Finnish time. The function $d(t)$ gives the local day of hour t; days change at midnight.

The study area is a 300×300 metre square around the fire station (VPK house) in Turku. It is divided into 10×10 metre squares. The model contains the

$$|\mathcal{C}| = 85$$

squares that lie within 5 metres of one of four hand-drawn walking routes. The crew works in larger units: 50×50 metre *blocks*, of which there are $|\mathcal{Z}| = 20$. The map $\zeta \colon \mathcal{C} \to \mathcal{Z}$ assigns each square to its block, and $\mathcal{C}_z = \zeta^{-1}(z)$ is the set of squares in block z.

2.3 Walking and weather

The following quantities are deterministic and known to every policy.

Base walking volume. The hourly base volume $\lambda_c > 0$ of square c is the sum of the assumed hourly volumes of the routes passing near it. For a block, $\lambda(z) = \sum_{c \in \mathcal{C}_z} \lambda_c$.

Daily rhythm. Let h_t be the clock hour at which hour t starts. Walking peaks in the morning and in the afternoon:

$$r_t = \Big(0.1 + 0.8\, e^{-((h_t-8)/2)^2} + 0.9\, e^{-((h_t-17)/3)^2}\Big) \cdot \begin{cases} 0.7 & \text{at weekends,} \\ 1 & \text{on weekdays.} \end{cases}$$

Weather. From the measured hourly means at Turku Artukainen (temperature T_t, snow depth s_t, precipitation q_t) we compute the weather factor

$$w_t = 0.7 + 1.2 \cdot \mathbf{1}\{-3 \le T_t \le 1\} + 0.6 \cdot \mathbf{1}\{s_t > 0\} + 0.5 \cdot \mathbf{1}\{q_t > 0,\ T_t < 1\},$$

which lies in $[0.7,\ 3.0]$. The base slip probability per passage is $\pi_t = p_0 w_t$ with $p_0 = 0.01$.

The weather factor is an experimental rule of the simulator and has not been calibrated against road-surface observations. It simply expresses that pavements are more slippery around zero degrees than in hard frost.

2.4 Hidden slipperiness

Every square has, on every day, its own *hidden slipperiness* $L_{c,d} \in [0, 1]$. It stands for things the weather does not tell: shade, slope, water dripping from eaves, or polished ice. The value $L = 1$ is the most slippery square possible and $L \approx 0$ is practically grippy.

We want a strongly skewed distribution: most squares are grippy, but a few are really slippery. A Beta distribution $\mathrm{Beta}(a, b)$ with a small parameter a has this shape. For exact computation the distribution is discretised.

Definition 1 (Slipperiness levels)**.** Let $a, b > 0$ and $G \in \mathbb{N}$. Let $B \sim \mathrm{Beta}(a, b)$ with distribution function F. Split $[0, 1]$ into G parts of equal probability at the points $q_j = F^{-1}(j/G)$, $j = 0, \dots, G$, and set

$$\ell_g = \mathbb{E}\big[B \mid q_{g-1} < B \le q_g\big], \qquad g = 1, \dots, G.$$

The slipperiness distribution is the uniform distribution on the levels, $\nu = \frac{1}{G}\sum_{g=1}^{G} \delta_{\ell_g}$, with mean $\mu = \sum_g \nu_g \ell_g$.

Lemma 1. *The levels of Definition 1 satisfy:*

(i) $\ell_g = G \cdot \frac{a}{a+b}\big(I_{q_g}(a+1, b) - I_{q_{g-1}}(a+1, b)\big)$*, where* I_x *is the regularised incomplete beta function;*

(ii) $0 < \ell_1 < \ell_2 < \cdots < \ell_G < 1$*;*

(iii) $\mu = a/(a+b)$*, i.e. the discretisation preserves the mean.*

Proof. The beta density satisfies $x f_{a,b}(x) = \frac{a}{a+b} f_{a+1,b}(x)$, because $B(a+1,b)/B(a,b) = a/(a+b)$. Each interval has probability $1/G$, so

$$\ell_g = G \int_{q_{g-1}}^{q_g} x f_{a,b}(x)\,dx = G \frac{a}{a+b} \int_{q_{g-1}}^{q_g} f_{a+1,b}(x)\,dx,$$

which is (i).

The density is positive on $(0,1)$. Hence $0 = q_0 < q_1 < \cdots < q_G = 1$, and the conditional mean of an interval lies strictly inside it: $q_{g-1} < \ell_g < q_g$. This gives (ii).

Part (iii) is the law of total expectation: $\mu = \sum_g \frac{1}{G}\,\mathbb{E}[B \mid \text{interval } g] = \mathbb{E}B = a/(a+b)$. □

Default values. The simulator uses $a = 1/4$, $b = 1$ and $G = 40$. When $b = 1$ the distribution function is simply $F(x) = x^a$, and the levels have the closed form

$$\ell_g = \frac{g^5 - (g-1)^5}{5\,G^4}.$$

The mean is $\mu = 0.2$, the most slippery level is $\ell_{40} \approx 0.951$ and the grippiest $\ell_1 \approx 7.8 \cdot 10^{-8}$. For the continuous distribution, $\mathbb{P}(B < 0.004) \approx 0.25$ and $\mathbb{P}(B > 0.5) \approx 0.16$.

> **Intuition** A quarter of the squares are practically grippy, and one in six is very slippery. Such a world rewards targeting: if you knew where the slippery squares were, gritting them would be many times as effective.

Slipperiness changes daily. The persistence parameter ρ says how likely yesterday's state is to remain.

Definition 2 (Dynamics of slipperiness)**.** Let $\rho \in [0,1]$. For every square c draw $L_{c,1} \sim \nu$. On the next day $L_{c,d+1} = L_{c,d}$ with probability ρ; otherwise $L_{c,d+1}$ is redrawn from ν. All draws are independent across squares and days.

Lemma 2. *For all c, d and $j \geq 0$ we have $L_{c,d} \sim \nu$ and*

$$\mathbb{E}[L_{c,d+j} \mid L_{c,1}, \ldots, L_{c,d}] = \mu + \rho^j (L_{c,d} - \mu).$$

Proof. The transition kernel is $P(\ell_g, \ell_h) = \rho\,\mathbf{1}\{g = h\} + (1-\rho)\nu_h$. Since $\nu P = \rho\nu + (1-\rho)\nu = \nu$ and the initial distribution is ν, induction gives $L_{c,d} \sim \nu$ for all d. By the Markov property, $\mathbb{E}[L_{c,d+1} \mid L_{c,\leq d}] = \rho L_{c,d} + (1-\rho)\mu$, and the claim follows by induction on j. □

The simulator's default is $\rho = 0$: every square's slipperiness is redrawn each day. Then the previous day's detections are of no use, and the service has to learn each day's situation from that day's detections.

2.5 Passages, slips and detections

All randomness is built on a single probability space so that independence statements are precise.

Definition 3 (Probability space)**.** The probability space $(\Omega, \mathcal{A}, \mathbb{P})$ carries the following mutually independent families of random variables:

(a) passages $N_{c,t} \sim \text{Poisson}(\lambda_c r_t)$, independent over all (c,t);

(b) the slipperiness draws of Definition 2;

(c) per-passage uniform variables $U_{c,t,i,j}$ with $i \in \mathbb{N}$ and $j \in \{1,2,3\}$;

(d) an optional randomisation V of a blind policy (Definition 5).

Write $\Psi_t = (N_{c,t}, U_{c,t,i,j})_{c,i,j}$ for the randomness of hour t and L for the whole slipperiness process. Then $L, V, \Psi_1, \Psi_2, \dots$ are independent.

Let $\gamma_{c,t} \in \{0,1\}$ indicate whether the block of square c is gritted in hour t; it is determined by the policy (Section 2.6). The remaining parameters of the model are

$e \in [0,1]$	effect of gritting,
φ	probability that a slip becomes a fall,
η	sensitivity of the detector,
ε	probability of a false alarm per passage.

The slip probability per passage is

$$p_{c,t} = \pi_t \, L_{c,d(t)} \, (1 - e\,\gamma_{c,t}). \tag{1}$$

We assume $\pi_t \ell_G < 1$ for all t; with the default values $\max_t \pi_t \ell_G \le 0.03 \cdot 0.951$.

Each passage $i = 1, \dots, N_{c,t}$ is treated separately:

- the passage leads to a slip if $U_{c,t,i,1} \le p_{c,t}$;
- a slip leads to a fall if $U_{c,t,i,2} \le \varphi$;
- the detector raises an alarm if a slip occurred and $U_{c,t,i,3} \le \eta$, or if no slip occurred and $U_{c,t,i,3} \le \varepsilon$.

The numbers of slips, falls and detections are denoted $S_{c,t}$, $F_{c,t}$ and $D_{c,t}$.

Lemma 3 (Distribution of detections and falls)**.** *Let $\mathcal{G}_t = \sigma(L, V, \Psi_1, \dots, \Psi_{t-1}, N_{\cdot,t})$, and let $\gamma_{\cdot,t}$ be $\sigma(L, V, \Psi_{<t})$-measurable. Then, conditionally on $\mathcal{G}_t$, the variables $D_{c,t}$, $c \in \mathcal{C}$, are independent and*

$$D_{c,t} \sim \text{Bin}\big(N_{c,t},\, \theta_t(L_{c,d(t)}, \gamma_{c,t})\big), \qquad \theta_t(\ell, \gamma) = \varepsilon + (\eta - \varepsilon)\, \pi_t\, \ell\, (1 - e\gamma).$$

Moreover $\mathbb{E}[F_{c,t} \mid \mathcal{G}_t] = \varphi\, N_{c,t}\, p_{c,t}$.

Proof. Conditionally on $\mathcal{G}_t$ the probability $p_{c,t}$ is known, and the variables $U_{c,t,\cdot,\cdot}$ are independent of $\mathcal{G}_t$. A single passage raises an alarm with probability $p_{c,t}\eta + (1 - p_{c,t})\varepsilon = \theta_t(L_{c,d(t)}, \gamma_{c,t})$,

and passages are independent of each other. Likewise a passage leads to a fall with probability $p_{c,t}\varphi$. □

Intuition The alarm probability θ is an *affine* function of slipperiness. When $\eta > \varepsilon$, a more slippery square produces more alarms per passage on average. This is the only channel through which the service learns anything about slipperiness.

Remark 1 (Implementation and common random numbers). The code does not draw every passage separately. It draws the totals S, F and D directly from binomial distributions by inversion, and every policy receives the same uniform random number. Conditionally on $(N_{c,t}, p_{c,t})$ the totals have the same distribution as above, so each policy's own distribution is exactly this model. The common random numbers couple the policies only so that their difference can be estimated with lower variance.

2.6 Gritting and the no-overlap rule

Definition 4 (Gritting rounds)**.** Let Δ, H and K be positive integers: the gritting interval, the duration of the effect and the number of blocks gritted per round. Rounds start at hours $\tau_k = 1 + (k-1)\Delta$, $k = 1, \dots, \kappa$, where $\kappa = \lceil T/\Delta \rceil$. The *effect window* of round k is

$$W_k = \{\tau_k, \dots, \tau_k + H - 1\} \cap \mathcal{T}.$$

In round k a set $A_k \subseteq \mathcal{Z}$ with $|A_k| = K$ is chosen, only among eligible blocks:

$$A_k \subseteq E_k := \mathcal{Z} \setminus \bigcup \{A_j : j < k,\ \tau_j + H > \tau_k\}.$$

This is the *no-overlap rule*: a block is not chosen while its previous gritting is still in effect. The gritting indicator is $\gamma_{c,t} = \mathbf{1}\{\exists k : \zeta(c) \in A_k,\ t \in W_k\}$.

Lemma 4. *Under the no-overlap rule, the effect windows of the same block are disjoint. Hence*

$$\gamma_{c,t} = \sum_k \mathbf{1}\{\zeta(c) \in A_k\}\, \mathbf{1}\{t \in W_k\}.$$

Proof. Let $j < k$ and $z \in A_j \cap A_k$. By eligibility $\tau_j + H \le \tau_k$, so $\max W_j < \tau_k = \min W_k$. The sum therefore has at most one nonzero term. □

2.7 Information, policies and objective

The service sees passages and vibration detections as they happen. Its information at the end of hour t is

$$\mathcal{F}_t = \sigma\big(N_{c,s},\ D_{c,s} : c \in \mathcal{C},\ s \le t\big),$$

and $\mathcal{F}_0$ is trivial.

Definition 5 (Policy)**.** A *policy* is a rule whose choice A_k is $\mathcal{F}_{\tau_k - 1}$-measurable and eligible: the decision is made before the round starts, using all information gathered so far. Because

detections depend on earlier gritting, the information depends on the policy; when needed we write $\mathcal{F}_t^\pi$.

- A policy is *blind* if A_k is measurable with respect to $\mathcal{B}_{\tau_k-1} \vee \sigma(V)$, where $\mathcal{B}_t = \sigma(N_{c,s} : s \le t)$. A blind policy may use the clock, the weather, passages and its own randomisation, but not vibration detections.

- The *oracle* has the information $\mathcal{H}_k = \sigma\big(L_{\cdot,d} : d \le d(\tau_k);\ \Psi_s : s < \tau_k\big)$. It knows the current day's slipperiness but not future randomness. For every policy, $\mathcal{F}^\pi_{\tau_k-1} \subseteq \mathcal{H}_k$.

Definition 6 (Objective)**.** The value of a policy π is the expected number of falls

$$J(\pi) = \mathbb{E}^\pi\Big[\sum_{c\in\mathcal{C}}\sum_{t\in\mathcal{T}} F_{c,t}\Big].$$

The simulator's column `expected_falls` is $\sum_{c,t}\varphi N_{c,t}p_{c,t}$. By Lemma 3 and the tower property its expectation is $J(\pi)$. It is therefore an unbiased estimator with lower variance than the drawn falls.

Symbol	Meaning
$c \in \mathcal{C}$, $z \in \mathcal{Z}$	square (10×10 m) and block (50×50 m)
t, $d(t)$	hour and its day
λ_c, r_t, π_t	base walking volume, daily rhythm, weather-driven base risk
$L_{c,d}$, ℓ_g, ν, μ	hidden slipperiness, its levels, distribution and mean
ρ	day-to-day persistence of slipperiness
N, S, F, D	passages, slips, falls, vibration detections
e, φ, η, ε	gritting effect, fall share, sensitivity, false alarms
Δ, H, K	gritting interval, effect duration, blocks per round
A_k, W_k, $\gamma_{c,t}$	round choice, effect window, gritting indicator
$\mathcal{F}_t$, $\mathcal{B}_t$, $\mathcal{H}_k$	information of the service, a blind policy and the oracle
$J(\pi)$	expected number of falls

Table 1: Main notation.

3 What the service knows: the beta-binomial posterior

The service does not see slipperiness, but it can infer it from detections with Bayes' rule. This section shows that the inference splits square by square and can be computed with a simple recursion.

Assume the policy is *deterministic*: the choice A_k is a deterministic function of the observation history. All policies in the code are, with ties broken by block number.

Proposition 1 (Filter)**.** *Conditionally on $\mathcal{F}_t$, the slipperiness chains $(L_{c,d})_d$, $c \in \mathcal{C}$, are independent. The filtering distribution of each square,*

$$w_{c,g}(t) = \mathbb{P}(L_{c,d(t)} = \ell_g \mid \mathcal{F}_t),$$

follows the recursion:

(i) Initially $w_c = \nu$. When the day changes, the distribution is predicted for the next day: $w_c \mapsto \rho\, w_c + (1-\rho)\nu$.

(ii) The detections of hour t update the distribution:

$$w_{c,g}(t) \;\propto\; w_{c,g}(t-1)\;\theta_t(\ell_g, \gamma_{c,t})^{D_{c,t}}\big(1-\theta_t(\ell_g,\gamma_{c,t})\big)^{N_{c,t}-D_{c,t}}.$$

The posterior mean of slipperiness is $m_c(t) = \sum_g w_{c,g}(t)\,\ell_g$.

Proof. Let $h = (n_{c,s}, \delta_{c,s})_{c\in\mathcal{C},\, s\le t}$ be an observation history and $l = (l_{c,d})$ a slipperiness path. Since the policy is deterministic, $\gamma_{c,s} = \gamma_{c,s}(h_{<s})$. Conditioning hour by hour with Lemma 3 gives

$$\mathbb{P}(N_{\le t} = n,\; D_{\le t} = \delta \mid L = l) = \prod_{s\le t}\prod_{c} \mathrm{Poisson}(n_{c,s};\lambda_c r_s)\binom{n_{c,s}}{\delta_{c,s}}\theta_s^{\delta_{c,s}}(1-\theta_s)^{n_{c,s}-\delta_{c,s}},$$

where $\theta_s = \theta_s\big(l_{c,d(s)}, \gamma_{c,s}(h_{<s})\big)$.

As a function of the path l this is a product of per-square factors. The prior is also a product, because the chains of different squares are independent. Hence the posterior is a product, and the posterior of each square is that of a hidden Markov model. Its transition kernel is P from Lemma 2, and the probability of an hour's observation is the binomial factor above. The forward algorithm gives (i) and (ii); the prediction in (i) is $wP = \rho w + (1-\rho)\nu$. □

In the code the update is the function `paivita_posteriori`. The simulator draws slipperiness from the same discrete distribution ν that the service uses as its prior, so the computed posterior is exact, not an approximation.

The name *beta-binomial* comes from the following special case, in which the prior is a continuous beta distribution and an alarm reveals slipperiness directly.

Proposition 2 (Conjugate special case)**.** *Let the prior be the continuous* $\mathrm{Beta}(a,b)$ *and let $\theta = \ell$, i.e. $\eta = 1$, $\varepsilon = 0$ and $\pi_s(1-e\gamma) = 1$. If during one day there are $n = \sum_s N_s$ passages and $y = \sum_s D_s$ detections, the posterior is* $\mathrm{Beta}(a+y,\; b+n-y)$*, and*

$$\mathbb{E}[L \mid \textit{data}] = \frac{a+y}{a+b+n} = \underbrace{\frac{a+b}{a+b+n}}_{\textit{weight of prior}}\cdot\frac{a}{a+b} \;+\; \underbrace{\frac{n}{a+b+n}}_{\textit{weight of data}}\cdot\frac{y}{n}.$$

Moreover, the marginal distribution of y given n is the beta-binomial distribution $\mathrm{BB}(n,a,b)$.

Proof. The posterior density is proportional to $\ell^{a-1}(1-\ell)^{b-1}\cdot\ell^{y}(1-\ell)^{n-y}$, which is the kernel of the $\mathrm{Beta}(a+y, b+n-y)$ density. The marginal is $\binom{n}{y}B(a+y,b+n-y)/B(a,b)$, which is the definition of the beta-binomial distribution. □

Intuition The prior $\mathrm{Beta}(a, b)$ is worth $a + b$ imaginary passages, a of which raised an alarm. The posterior mean is a weighted average: with little data it stays near the prior mean, with much data it approaches the observed share y/n. A single alarm from a single passage does not make a square slippery.
In the simulator θ is an affine function of slipperiness, so the posterior is not of beta form. It is therefore computed exactly on the grid of levels $\ell_1, \ldots, \ell_G$. The check `tarkista_tammikuu.R` verifies that the grid computation reproduces Proposition 2 when $G = 4000$.

4 Decomposing the value of gritting

The number of falls in a month is a complicated random variable: every gritting affects future slips, detections and decisions. This section shows that its expectation nevertheless splits neatly into one term per gritting round.

Define the *risk mass* of block z in the effect window of round k and its deterministic counterpart, the *exposure*:

$$R_{z,k} = \sum_{c \in \mathcal{C}_z} \sum_{t \in W_k} N_{c,t}\, \pi_t\, L_{c,d(t)}, \qquad \Lambda_{z,k} = \lambda(z)\, \kappa_k, \quad \kappa_k = \sum_{t \in W_k} r_t \pi_t.$$

The risk mass is the amount of slip risk that gritting could remove. It does not depend on the policy, because it is a function of passages and slipperiness only.

Proposition 3 (Decomposition)**.** *Let $\mathcal{G}_k$ be any σ-algebra with respect to which A_k is measurable. For every policy that respects the no-overlap rule,*

$$J(\pi) = \mathsf{B} - e\varphi \sum_{k=1}^{\kappa} \mathbb{E}^{\pi}\Big[\sum_{z \in A_k} \mathbb{E}[R_{z,k} \mid \mathcal{G}_k] \Big], \qquad \mathsf{B} = \varphi\mu \sum_{c,t} \lambda_c r_t \pi_t = J(\texttt{no-grit}).$$

Proof. Step 1. By Lemma 3 and the tower property, $\mathbb{E}F_{c,t} = \varphi\, \mathbb{E}\big[N_{c,t}\, \pi_t L_{c,d(t)} (1 - e^{\gamma_{c,t}})\big]$.

Step 2. By Lemma 4 the gritted part can be grouped by round:

$$\sum_{c,t} N_{c,t}\pi_t L_{c,d(t)} \gamma_{c,t} = \sum_{k} \sum_{z \in A_k} \sum_{c \in \mathcal{C}_z} \sum_{t \in W_k} N_{c,t}\pi_t L_{c,d(t)} = \sum_{k} \sum_{z \in A_k} R_{z,k}.$$

Step 3. Since $N \perp L$ and $L_{c,d} \sim \nu$, $\mathbb{E}[N_{c,t}\pi_t L_{c,d(t)}] = \lambda_c r_t \pi_t \mu$, which gives B.

Step 4. Finally $\mathbb{E}[\mathbf{1}\{z \in A_k\} R_{z,k}] = \mathbb{E}\big[\mathbf{1}\{z \in A_k\}\, \mathbb{E}[R_{z,k} \mid \mathcal{G}_k]\big]$, because $\mathbf{1}\{z \in A_k\}$ is $\mathcal{G}_k$-measurable and $R_{z,k} \geq 0$ is integrable. □

Intuition The no-overlap rule guarantees that no square-hour ever benefits from two grittings. The value of a policy is then a sum over rounds: how much risk mass the chosen blocks contain *in the light of the information available when choosing*. The best policy is the one that can pick the blocks with the largest risk mass.

The next lemma says what "the information available when choosing" amounts to for the different policies.

Lemma 5 (Conditional risk mass)**.** *Let $\mathcal{G} \subseteq \sigma(L, V, \Psi_{<\tau_k})$ and $m_{c,d} = \mathbb{E}[L_{c,d} \mid \mathcal{G}]$. Then*

$$\mathbb{E}[R_{z,k} \mid \mathcal{G}] = \sum_{c \in \mathcal{C}_z} \lambda_c \sum_{t \in W_k} r_t \pi_t \, m_{c,d(t)}.$$

In particular:

(i) for the service ($\mathcal{G} = \mathcal{F}^{\pi}_{\tau_k - 1}$), $m_{c,d}$ is the posterior mean of Proposition 1;

(ii) for a blind policy ($\mathcal{G} = \mathcal{B}_{\tau_k - 1} \vee \sigma(V)$), $m_{c,d} = \mu$, so $\mathbb{E}[R_{z,k} \mid \mathcal{G}] = \mu \Lambda_{z,k}$;

(iii) for the oracle ($\mathcal{G} = \mathcal{H}_k$), $m_{c,d} = \mu + \rho^{d - d(\tau_k)}(L_{c,d(\tau_k)} - \mu)$ for $d \geq d(\tau_k)$.

Proof. For $t \geq \tau_k$ the passage count $N_{c,t}$ belongs to the randomness Ψ_t and is therefore independent of $\sigma(L, V, \Psi_{<\tau_k})$, which contains $\mathcal{G} \vee \sigma(L)$. Hence $\mathbb{E}[N_{c,t} L_{c,d(t)} \mid \mathcal{G}] = \mathbb{E}[N_{c,t}] \, \mathbb{E}[L_{c,d(t)} \mid \mathcal{G}]$.

In (ii), L is independent of N and V, so $\mathbb{E}[L_{c,d} \mid \mathcal{G}] = \mathbb{E} L_{c,d} = \mu$. Part (iii) follows from Lemma 2, because the draws of future days are independent of $\mathcal{H}_k$. □

5 Main results

5.1 Assumptions and policies

Two assumptions make the rounds independent selection problems.

Assumption 1 (No carry-over)**.** $H \leq \Delta$: the effect of gritting ends no later than the start of the next round.

Assumption 2 (Window within one day)**.** For every k and every $t \in W_k$ we have $d(t) = d(\tau_k)$.

Under Assumption 1, $\tau_j + H \leq \tau_k$ whenever $j < k$. Then $E_k = \mathcal{Z}$: every set of K blocks is eligible, and earlier choices do not restrict later ones.

The simulator's defaults are $\Delta = H = 6$ and $K = 2$. Since 6 divides 24, the first round starts at midnight and January has no daylight-saving change, Assumption 2 also holds.

Definition 7 (Policies of the simulator)**.** Write $\mathcal{A}_K = \{A \subseteq \mathcal{Z} : |A| = K\}$.

- `no-grit`: no gritting, $\gamma \equiv 0$.
- `rotation`: the next K eligible blocks in cyclic order.
- `busiest`: $A_k \in \arg\max_{A \in \mathcal{A}_K} \sum_{z \in A} \Lambda_{z,k}$, i.e. the K blocks with the largest $\lambda(z)$.
- `pito`: $A_k \in \arg\max_{A \in \mathcal{A}_K} \sum_{z \in A} \hat{S}_{z,k}$, where $\hat{S}_{z,k} = \mathbb{E}[R_{z,k} \mid \mathcal{F}^{\texttt{pito}}_{\tau_k - 1}]$.
- `oracle`: $A_k \in \arg\max_{A \in \mathcal{A}_K} \sum_{z \in A} \mathbb{E}[R_{z,k} \mid \mathcal{H}_k]$.

5.2 The best blind policy

Theorem 1 (Best blind policy)**.** *Under Assumption 1, every blind policy β satisfies*

$$J(\beta) \;\geq\; J^*_{\text{blind}} := \mathsf{B} - e\varphi\mu \sum_k \max_{A\in\mathcal{A}_K} \sum_{z\in A} \Lambda_{z,k} \;=\; J(\texttt{busiest}).$$

Proof. Take $\mathcal{G}_k = \mathcal{B}_{\tau_k-1} \vee \sigma(V)$ in Proposition 3. By Lemma 5 (ii),

$$J(\beta) = \mathsf{B} - e\varphi\mu \sum_k \mathbb{E}\Big[\sum_{z\in A_k} \Lambda_{z,k}\Big].$$

Since $A_k \in \mathcal{A}_K$, the sum is at most the maximum. The policy `busiest` attains the maximum in every round. Since $\Lambda_{z,k} = \lambda(z)\kappa_k$ and $\kappa_k > 0$, the maximising set consists of the K blocks with the largest $\lambda(z)$. □

Corollary 1. *$J(\texttt{rotation}) \geq J(\texttt{busiest})$. The same holds for every rule that chooses blocks by clock, weather or observed passages.*

> **Intuition** Without vibration detections, the best estimate of every square's slipperiness is the mean μ. The value of gritting is then proportional to exposure alone, and the best blind rule is to grit the busiest blocks. Observed passages say nothing about slipperiness, because passages and slipperiness are independent in the model.

5.3 The value of vibration detections

Theorem 2 (Value of detections)**.** *Under Assumption 1, $J(\texttt{pito}) \leq J^*_{\text{blind}}$, so $J(\texttt{pito}) \leq J(\beta)$ for every blind policy β. More precisely,*

$$J(\texttt{busiest}) - J(\texttt{pito}) = e\varphi \sum_k \mathbb{E}\Big[\max_{A\in\mathcal{A}_K} \sum_{z\in A} \hat{S}_{z,k} - \sum_{z\in A_k^{\texttt{busiest}}} \hat{S}_{z,k}\Big] \;\geq\; 0, \tag{2}$$

where every term inside the expectation is pointwise non-negative.

Proof. Take $\mathcal{G}_k = \mathcal{F}^{\texttt{pito}}_{\tau_k-1}$ in Proposition 3. Since `pito` maximises,

$$J(\texttt{pito}) = \mathsf{B} - e\varphi \sum_k \mathbb{E}\Big[\max_{A\in\mathcal{A}_K} \sum_{z\in A} \hat{S}_{z,k}\Big].$$

The set $A_k^{\texttt{busiest}}$ is deterministic and $\mathbb{E}\hat{S}_{z,k} = \mathbb{E}R_{z,k} = \mu\Lambda_{z,k}$. Hence

$$\mathbb{E}\Big[\sum_{z\in A_k^{\texttt{busiest}}} \hat{S}_{z,k}\Big] = \mu \max_{A\in\mathcal{A}_K} \sum_{z\in A} \Lambda_{z,k}.$$

Subtracting gives (2). Each term is non-negative because the maximum is at least the value of the sum on any set, in particular on $A_k^{\texttt{busiest}}$. □

Intuition A service that uses detections can always pick the same set as the best blind rule. It deviates only when the posterior promises more risk mass. The posterior mean equals the prior on average, so the deviations cannot make the choice worse on average. This is the basic value-of-information result $\mathbb{E}\max \geq \max \mathbb{E}$.

5.4 The oracle

Theorem 3 (Oracle)**.** *Under Assumption 1, $J(\texttt{oracle}) \leq J(\pi)$ for every policy π. In particular $J(\texttt{oracle}) \leq J(\texttt{pito})$.*

Proof. Since $\mathcal{F}^{\pi}_{\tau_k - 1} \subseteq \mathcal{H}_k$, the set A^{π}_k is $\mathcal{H}_k$-measurable. By Proposition 3,

$$J(\pi) = \mathsf{B} - e\varphi \sum_k \mathbb{E}\Big[\sum_{z \in A^{\pi}_k} \mathbb{E}[R_{z,k} \mid \mathcal{H}_k] \Big].$$

The inner sum is at most $\max_{A \in \mathcal{A}_K} \sum_{z \in A} \mathbb{E}[R_{z,k} \mid \mathcal{H}_k]$, which the oracle attains. □

5.5 When the gain is strict

Theorem 2 says that detections do no harm. The next theorem says when they genuinely help.

Theorem 4 (Strict gain)**.** *Assume Assumptions 1 and 2 and the following conditions.*

(C1) Discrimination: $0 < \varepsilon < \eta \leq 1$, $0 < e < 1$ *and* $\varphi > 0$.

(C2) Variation: $G \geq 2$.

(C3) Room to choose: $K < |\mathcal{Z}|$, $\Delta < 24$, *and there are blocks* $z^* \in A_2^{\texttt{busiest}}$ *and* $z' \notin A_2^{\texttt{busiest}}$ *with* $\ell_G\, \lambda(z') > \ell_1\, \lambda(z^*)$.

Then $J(\texttt{pito}) < J(\texttt{busiest})$.

Proof. The terms of (2) are non-negative and $e\varphi > 0$. It therefore suffices to show that in round $k = 2$, with positive probability,

$$\max_{A \in \mathcal{A}_K} \sum_{z \in A} \hat{S}_{z,2} \;>\; \sum_{z \in A_2^{\texttt{busiest}}} \hat{S}_{z,2}. \tag{3}$$

Step 1: a swap is enough. Since $\Delta < 24$, round $\tau_2 = 1 + \Delta$ falls on the first day. By Assumption 2 and Lemma 5,

$$\hat{S}_{z,2} = \kappa_2 \sum_{c \in \mathcal{C}_z} \lambda_c m_c, \qquad m_c = \mathbb{E}[L_{c,1} \mid \mathcal{F}_\Delta], \quad \kappa_2 > 0.$$

If $\sum_{c \in \mathcal{C}_{z'}} \lambda_c m_c > \sum_{c \in \mathcal{C}_{z^*}} \lambda_c m_c$, the set $(A_2^{\texttt{busiest}} \setminus \{z^*\}) \cup \{z'\}$ gives a strictly larger sum, and (3) holds.

Step 2: extreme but possible data. Let $n \in \mathbb{N}$ and let E_n be the event that every square $c \in \mathcal{C}_{z'} \cup \mathcal{C}_{z^*}$ has $N_{c,1} = n$ and $N_{c,s} = 0$ for $2 \le s \le \Delta$, while $D_{c,1} = n$ on the squares of z' and $D_{c,1} = 0$ on the squares of z^*. Since $\lambda_c r_s > 0$, and by (C1) and $\pi_t \ell_G < 1$ we have $0 < \varepsilon \le \theta_1(\ell, \gamma) < 1$, Lemma 3 gives $\mathbb{P}(E_n) > 0$.

Step 3: the posterior concentrates. On E_n, Proposition 1 gives on the first day

$$w_{c,g} = \frac{\theta_1(\ell_g, \gamma_c)^n}{\sum_h \theta_1(\ell_h, \gamma_c)^n} \quad (c \in \mathcal{C}_{z'}), \qquad w_{c,g} = \frac{(1 - \theta_1(\ell_g, \gamma_c))^n}{\sum_h (1 - \theta_1(\ell_h, \gamma_c))^n} \quad (c \in \mathcal{C}_{z^*}),$$

because the prior is uniform and the hours $s \ge 2$ contain no passages. By (C1), (C2) and Lemma 1 (ii), $\theta_1(\ell, \gamma)$ is strictly increasing in ℓ for both $\gamma \in \{0, 1\}$. As $n \to \infty$ the weights concentrate on level G in block z' and on level 1 in block z^*. Hence

$$\sum_{c \in \mathcal{C}_{z'}} \lambda_c m_c - \sum_{c \in \mathcal{C}_{z^*}} \lambda_c m_c \;\longrightarrow\; \ell_G \lambda(z') - \ell_1 \lambda(z^*) \;>\; 0.$$

Step 4: conclusion. The limit does not depend on other data, because each square's posterior depends only on its own observations. Choose n large enough for the difference to be positive on E_n. Then (3) holds with probability at least $\mathbb{P}(E_n) > 0$. □

The simulator's defaults satisfy the conditions: $\varepsilon = 0.0002 < \eta = 0.8$, $e = 0.8$, $\varphi = 0.02$, $G = 40$, $K = 2 < 20$ and $\Delta = 6$. Moreover $\ell_1/\ell_G \approx 8 \cdot 10^{-8}$, while the smallest $\lambda(z)$ is 16 and the largest 157 passages per hour. Condition (C3) therefore holds for every pair of blocks.

> **Intuition** The proof builds one clear-cut situation: in the early morning a quiet block produces nothing but alarms and a busy block none at all. The best blind rule still grits the busy block, but the service knows better. Because the situation has positive probability, the expected gain is strictly positive.

5.6 When there is no gain

Theorem 5 (Necessary conditions)**.** *Under Assumption 1:*

(i) if $\eta = \varepsilon$, then $J(\texttt{pito}) = J(\texttt{busiest})$;

(ii) if $G = 1$, then $J(\texttt{oracle}) = J(\texttt{pito}) = J(\texttt{busiest})$;

(iii) if $e = 0$ or $\varphi = 0$, then $J(\pi) = \mathsf{B}$ for every policy.

Proof. (i) When $\eta = \varepsilon$, the probability θ_t does not depend on slipperiness, so the update of Proposition 1 leaves the distribution unchanged. Then $w(t) = \nu$ and $m_c = \mu$, so $\hat{S}_{z,k} = \mu \Lambda_{z,k}$ is deterministic and the difference in (2) is zero.

(ii) When $G = 1$, slipperiness is the constant μ, and all three policies score blocks by $\mu \Lambda_{z,k}$.

(iii) The coefficient of the gain term in Proposition 3 is $e\varphi = 0$. □

5.7 Summary and remarks

Corollary 2 (Ordering)**.** *Under Assumption 1,*

$$J(\texttt{oracle}) \le J(\texttt{pito}) \le J(\texttt{busiest}) \le J(\texttt{rotation}) \le J(\texttt{no-grit}) = \mathsf{B}.$$

Under the conditions of Theorem 4, the second inequality is strict.

Remark 2 (Longer effect). When $H > \Delta$, Lemma 4 and Proposition 3 still hold thanks to the no-overlap rule. The eligible sets E_k, however, depend on earlier choices, so the proofs do not carry over as such. A choice based on detections may, for example, take a block out of use at a moment when a blind rule would have used it. The code allows this setting, but its results in Table 5 are empirical only.

Remark 3 (Greediness). `pito` maximises the conditional gain of each round separately. The theorems do not claim it is optimal among all policies that use detections. Gritting weakens a block's future detections, since θ shrinks by the factor $(1 - e)$, so a more exploratory policy could do better. The proof of Theorem 2 nevertheless applies verbatim to any policy that chooses $\arg\max_A \sum_{z \in A} \mathbb{E}[R_{z,k} \mid \mathcal{G}_k]$ for some σ-algebra $\mathcal{G}_k \subseteq \mathcal{H}_k$.

6 Correspondence between code and model

The code is in Finnish. Table 2 shows where each symbol lives in the code, and Table 3 maps the policy names of this paper to the code's identifiers. The files are in `If/labra/v0`.

Symbol	In the code	Default
a, b, G	`liukkaus_a`, `liukkaus_b`, `liukkaustasoja`	0.25; 1; 40
ρ	`pysyvyys`	0
p_0	`liukastuminen_p`	0.01
φ	`kaatuminen_liukastumisesta_p`	0.02
η, ε	`tunnistus_p`, `virhehavainto_p`	0.8; 0.0002
Δ, H, K	`huoltovali_h`, `huollon_kesto_h`, `huoltokohteita`	6; 6; 2
e	`huollon_teho`	0.8
ℓ_g	`liukkaustasot()`	Lemma 1
$w_{c,g}$	`exp(logw)`, `paivita_posteriori()`	Proposition 1
w_t, r_t	`saakerroin()`, `rytmi`	Section 2

Table 2: Notation of the paper and variables of the simulator (`tammikuu_asetukset.R`, `R/simuloi_tammikuu.R`).

This paper	Code identifier	Meaning
`no-grit`	`ei_huoltoa`	no gritting
`rotation`	`aikataulu`	fixed rotation
`busiest`	`kulku`	busiest blocks, best blind policy
`pito`	`tuote`	detection-based targeting (the product)
`oracle`	`oraakkeli`	knows the hidden slipperiness

Table 3: Policy names in this paper and in the code.

The code scores block z in round k by

$$\text{score}(z) = \sum_{c \in \mathcal{C}_z} \lambda_c(\mu A_0 + (m_c - \mu)A_1), \qquad A_0 = \sum_{t \in W_k} r_t, \quad A_1 = \sum_{t \in W_k} r_t \, \rho^{d(t)-d(\tau_k)}.$$

Here $m_c = \mu$ for `busiest`, the posterior mean for `pito`, and $L_{c,d(\tau_k)}$ for `oracle`. The weather factor is left out so that the decision does not need future weather.

Proposition 4. *Under Assumptions 1 and 2, the code's* `busiest`, `pito` *and* `oracle` *choose an argmax set of Definition 7.*

Proof. Under Assumption 2, $A_1 = A_0$, so $\text{score}(z) = A_0 \sum_{c \in \mathcal{C}_z} \lambda_c m_c$. By Lemma 5 the exact value is $\kappa_k \sum_{c \in \mathcal{C}_z} \lambda_c m_c$ with the same conditional mean m_c, and by Proposition 1 the code's posterior is exact. Since $A_0 > 0$ and $\kappa_k > 0$, the scores are positive multiples of each other, so their argmax sets coincide. Under Assumption 1 the eligibility constraint removes no set. □

The checks `tarkista_tammikuu.R` and `tarkista_hyoty.R` test:

- the mean of the discretisation (Lemma 1);
- that the filter agrees with a directly computed posterior (Proposition 1);
- the conjugate special case (Proposition 2);
- the no-overlap rule, also when $H > \Delta$;
- that future weather does not affect past decisions;
- the case $G = 1$ (Theorem 5 (ii)).

7 Numerical results

The theorems give the ordering of the results but not the size of the differences. The differences are estimated by simulating January 2026 with many random seeds. Each seed has its own hidden slipperiness and its own passages, and all policies use the same random numbers (Remark 1).

The measure is `expected_falls`. Confidence intervals are $\bar{x} \pm 1.96\, s/\sqrt{n}$ over seeds, and differences are paired. The results are computed with `Rscript paperi/laske_taulukot.R`; a Finnish version of this paper is `paperi_fi.pdf`.

Interpretation. The ordering in the tables matches the theorems. With the default settings, `pito` beat `busiest` in 193 of 200 seeds, and the confidence interval [0.38, 0.44] of the paired difference excludes zero (Theorem 4). In the settings $G = 1$ and $\eta = \varepsilon$ the difference was exactly zero for every seed, as Theorem 5 predicts.

Policy π	$\hat{J}(\pi)$ [95 % CI]	Reduction, %	$\hat{J}(\texttt{busiest}) - \hat{J}(\pi)$ [95 % CI]
`no-grit`	25.05 [24.96, 25.15]	0.0	−4.13 [−4.17, −4.09]
`rotation`	22.98 [22.89, 23.07]	8.3	−2.06 [−2.10, −2.02]
`busiest`	20.92 [20.83, 21.01]	16.5	–
`pito`	20.51 [20.43, 20.60]	18.1	0.41 [0.38, 0.44]
`oracle`	19.17 [19.09, 19.25]	23.5	1.75 [1.72, 1.79]

Table 4: Expected falls in January with the default settings (200 seeds). The reduction is relative to `no-grit`.

Setting	n	`no-grit`	`rot.`	`busiest`	`pito`	`oracle`	Gap [95 % CI]	> 0	Share
Default ($\rho = 0$)	200	25.1	23.0	20.9	20.5	19.2	0.41 [0.38, 0.44]	193	23 %
$\rho = 0.5$	100	25.0	23.0	20.8	20.3	19.1	0.54 [0.47, 0.61]	96	31 %
$\rho = 0.9$	100	25.0	23.0	20.9	19.8	19.1	1.12 [0.95, 1.29]	92	62 %
No variation ($G = 1$)	100	25.0	23.0	20.9	20.9	20.9	0.00 [0.00, 0.00]	0	–
No discrimination ($\eta = \varepsilon$)	100	25.1	23.0	20.9	20.9	19.2	0.00 [0.00, 0.00]	0	0 %
$H = 12 > \Delta$	100	25.1	21.1	17.3	17.0	15.3	0.29 [0.25, 0.32]	96	14 %

Table 5: Sensitivity. Gap is $\hat{J}(\texttt{busiest}) - \hat{J}(\texttt{pito})$, and `rot.` is `rotation`. Column >0 counts the seeds in which `pito` beat `busiest`. Share is $(\hat{J}(\texttt{busiest}) - \hat{J}(\texttt{pito}))/(\hat{J}(\texttt{busiest}) - \hat{J}(\texttt{oracle}))$, the part of the advantage of perfect information that the detections recover.

Most of `pito`'s advantage over `rotation` comes from taking walking volumes into account: `busiest` removes about 2.1 expected falls and the vibration detections about 0.4 more. The detections recover roughly a quarter of the advantage of perfect information. With the default $\rho = 0$ the slipperiness is redrawn at midnight, so the 00:00 round has no information about the new day and the 06:00 round only a quiet night's worth. Mainly the 12:00 and 18:00 rounds are informed. When slipperiness persists from day to day ($\rho = 0.9$), information accumulates and the share rises to about 60 per cent.

Even the oracle removes only about 24 per cent of the falls, because the crew covers 2 of the 20 blocks at a time. The setting $H = 12$ lies outside Assumption 1. There `pito` was still better on average, but the theorems do not guarantee it (Remark 2).

The numbers are simulated falls in a 300×300 metre area over one month. They are not insurance claims or real events in Turku.

8 Limitations and further work

The results are only as good as the model. The main limitations are these.

The model is known. The service's posterior uses the simulator's true parameters. A misspecified prior or a wrong detector accuracy may weaken the result, and Theorem 2 does not cover that case.

Every passage is observed. The share of app users has been removed from the model. A partial sample would reduce the number of observations and thus the information. The theorems still hold if the unobserved part is independent of slipperiness.

Independence. Passages, slips and alarms are conditionally independent. The model has no people stepping aside, no groups and no repeated passages by the same person.

Slipperiness is constant within a day, and squares are independent. Spatial correlation would probably improve the estimate, because neighbouring squares could borrow information from each other. This is a natural next step.

The weather factor, the gritting effect and the fall share are assumptions. They have not been calibrated. Falls are not insurance claims, and converting them to euros needs a separate model.

Routes and walking volumes are drawn by hand and are not measured data.